PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Acupuncture for biliary colic: a systematic review protocol
AUTHORS	Sun, Ning; Zuo, Wenwei; Zhou, Yuanfang; Cheng, Ying; Cheng, Shirui; Zhou, Jun; Xu, Guixing; Huang, Liuyang; Liang, Fan-rong; Sun, Ruirui

VERSION 1 – REVIEW

REVIEWER	Robin Prescott Centre for Population Health Sciences, Usher Institute, University of Edinburgh, Scotland
REVIEW RETURNED	06-Aug-2020

GENERAL COMMENTS	I have been asked to look at this protocol as a statistical referee. In general, I like this protocol. For the most part, the design is fairly standard, and none the worse for that. There are, though, some detailed points for the authors to consider. My first question for the authors is a non-statistical point. I understand that biliary colic can occur in association with gallstones and it can also occur after cholecystectomy, when it is described as the postcholecystectomy syndrome. Are the authors intending to study biliary colic in both situations? If so, there will be overlap with the following protocol: Yin Z, Cheng Y, Xiao Q, et al. Acupuncture for the postcholecystectomy syndrome: Protocol for a systematic review and meta-analysis. Medicine (Baltimore). 2019;98(32):e16769. doi:10.1097/MD.0000000000016769. The protocol should be modified to provide clarity on this point and, if relevant, note the overlap with another systematic review. I found myself slightly confused when reading the section on 'Dealing with missing data'. I would understand these comments if the authors were planning to conduct an analysis using individual participant data. However, the previous section clarifies that this is not the case and the analysis will be based on the aggregated outcome data. Therefore, I have difficulty in envisaging any situation where the last observation carried forward imputation method could ever be used. Indeed, I would doubt that any form of imputation would be relevant to, or desirable in, a meta analysis based on aggregated outcome data. If the authors can envisage such a situation, it would aid comprehension if an example of when imputation might be used was provided. I certainly support the authors making efforts to obtain any information that might be needed for the systematic review, and which was unavailable in the publication, but I would need convincing as to the wisdom of imputing such information.

	I have a fundamental difference of opinion with the authors about the contents of the 'Assessment of heterogeneity' section of the protocol. The fixed effect model has the assumption that there is one true effect size which is common to all studies. Any differences that are seen between studies are assumed to be due solely to sampling error. In contrast, the random effects model allows for the possibility that the true effect sizes might differ between one study and another. The latter model is far more realistic and I feel that there are only rare occasions when a fixed effects model might be appropriate. I do not see any of those occasions occurring in the situation of a meta analysis of published papers. Why should the effect size be identical in studies with different populations, with different staff administering the treatments, and where the interventions might not even be identical? Thus I would urge the authors to modify their protocol to use a random effects model in all circumstances. Reporting the value of I-squared is useful as a summary measure of between study variability, but the only time I feel it should change the analysis is when its values are very large. In those circumstances, I suggest that the authors plan to investigate possible causes of the large degree of heterogeneity. I note that the authors hint at that in the 'Data synthesis' section. and also in the 'Subgroup analysis' section. There would be a benefit to the protocol if these 3 sections were harmonised to avoid repetition when possible, with cross-referencing across sections when necessary. At the start of the section on 'Assessment of heterogeneity', the term forest plot survey is used, and the authors specify a significance level of 0.1 when investigating statistical heterogeneity. I think the authors simply need to say that forest plots will be used to present effect sizes and corresponding 95% confidence limits for each study. The term 'survey' is unsuitable. In view of my suggestion that random effects models should always be used, I think that a significance test for heterogeneity is futile. The random effects model already assumes that there could be different effect sizes and the Null hypothesis of no differences in effect sizes is therefore irrelevant. The authors have clearly thought about the search strategy and I do not have the expertise to comment, other than to mention a typo in Table 1 where, in step 16, 'mamual' should be replaced by 'manual'. I notice, though, that the authors have not included acupressure in their search. Would such a method fall within the scope of this systematic review? Unless the authors have already done so, I would urge them to consult an experienced librarian about the search strategy, as librarians are the experts. Comparing the search strategy in this protocol to that in the protocol of Yin et al., I see that the search for acupuncture studies is far more detailed in the latter study.

REVIEWER	Guang Ji Shanghai University of Traditional Chinese Medicine, Shanghai
REVIEW RETURNED	06-Sep-2020

GENERAL COMMENTS	This is a well-designed protocol for a systematic review. I only have a few suggestions for authors. 1. The RevMan software has been updated to version 5.4. In future data analysis, this version should be considered.

	2. Authors mentioned acupuncture details will be corrected based on STRICTA statement. I suggest author list the detailed information. 3. Publication bias will be evaluated through a funnel plot according to manuscript. I suggest Egger's test could be used to assess the imbalance. 4. In addition, meta-regression could be considered for exploring the potential resources of heterogeneity.

VERSION 1 – AUTHOR RESPONSE

Responses to Reviewer 1:
I have been asked to look at this protocol as a statistical referee. In general, I like this protocol. For the most part, the design is fairly standard, and none the worse for that. There are, though, some detailed points for the authors to consider.
Re: We gratefully thanks for the precious time the reviewer spent making constructive remarks.

1. My first question for the authors is a non-statistical point. I understand that biliary colic can occur in association with gallstones and it can also occur after cholecystectomy, when it is described as the postcholecystectomy syndrome. Are the authors intending to study biliary colic in both situations? If so, there will be overlap with the following protocol: Yin Z, Cheng Y, Xiao Q, et al. Acupuncture for the postcholecystectomy syndrome: Protocol for a systematic review and meta-analysis. Medicine (Baltimore). 2019;98(32):e16769. doi:10.1097/MD.0000000000016769. The protocol should be modified to provide clarity on this point and, if relevant, note the overlap with another systematic review.
Re: Thank you for the reviewer's rigorous consideration. In this study, we intend to study biliary colic with gallstones, not including postcholecystectomy syndrome. I have amended it in the protocol to clarify this point.

① Description of the condition
Biliary colic (BC) is an episodic attack of severe pain for at least 20–30 minutes in the right upper abdominal quadrant, which even radiates to the right-back or shoulder.1 BC occurs within a couple of hours after eating a meal. It is the most common manifestation of gallstones.2 BC often accompanying by nausea, vomiting or fever, affects at least one-third of the adult population with gallstones.3 BC may also occur after cholecystectomy, which is described as postcholecystectomy syndrome (PCS).
② Why it is important to perform this review
Nowadays, the number of acupuncture clinical reports for BC has increased gradually in recent years.15,20 So far, only one systematic review (SR) published in 2020 resulted that acupuncture plus conventional medicine (CM) with single CM did not show statistical differences in reduction of pain for patients with PCS. 21 However, what is the effect of acupuncture on BC patients without cholecystectomy? Therefore, a lack of SR to address the following pivotal questions: ① whether acupuncture an effective therapy for BC with gallstones; ② whether acupuncture an effective adjunctive therapy to CM for BC with gallstones. Thus, there is a chance for us to conceive a comprehensive review to determine the efficacy and safety of acupuncture for BC.
21 Yin Z, Xiao Q, Xu G, et al. Acupuncture for the Postcholecystectomy Syndrome: A Systematic Review and Meta-Analysis. Evid Based Complement Alternat Med. 2020;2020:7509481.
③ Types of participants
All people presenting with BC with gallstones according to the definitions and diagnostic work-up in the individual trials.

2. I found myself slightly confused when reading the section on ‘Dealing with missing data’. I would understand these comments if the authors were planning to conduct an analysis using individual participant data. However, the previous section clarifies that this is not the case and the analysis will be based on the aggregated outcome data. Therefore, I have difficulty in envisaging any situation where the last observation carried forward imputation method could ever be used. Indeed, I would doubt that any form of imputation would be relevant to, or desirable in, a meta analysis based on aggregated outcome data. If the authors can envisage such a situation, it would aid comprehension if an example of when imputation might be used was provided. I certainly support the authors making efforts to obtain any information that might be needed for the systematic review, and which was unavailable in the publication, but I would need convincing as to the wisdom of imputing such information.
Re: We appreciate and agree with the reviewer’s advice. As you can see in the paper, we do plan to conduct an analysis based on the aggregated outcome data. Sorry for the unclear expression and insufficient data processing skills. We have modified the strategies of dealing the missing data according to your opinion.

Dealing with missing data
Whether the data were intentionally missing or ‘randomly’ missing can be determined by contacting the corresponding author or author concerned. After the investigator concludes that the data are missing randomly, the available data will be analyzed.25 If not, it is necessary to apply to missing data to the trial’s original investigator of the data or the contact listed in the trial registry. In the event of no reaction from the authors or contact person, we will apply the following strategies to address missing data:
1. In the included studies, if intention to-treat (ITT) analyses were performed,26 we will use ITT data as the first choice, not missing data.
2. For continuous missing outcome data, if the medians, p values or confidence intervals were reported. we will try to re-calculate mean difference and standard deviation values as the first choice. At the same time, if feasible, we will conduct a sensitivity analyses to address the potential impact of missing data.27 The potential impact of the effect of missing data on the final findings of the review will be addressed in the discussion.

3. I have a fundamental difference of opinion with the authors about the contents of the ‘Assessment of heterogeneity’ section of the protocol. The fixed effect model has the assumption that there is one true effect size which is common to all studies. Any differences that are seen between studies are assumed to be due solely to sampling error. In contrast, the random effects model allows for the possibility that the true effect sizes might differ between one study and another. The latter model is far more realistic and I feel that there are only rare occasions when a fixed effects model might be appropriate. I do not see any of those occasions occurring in the situation of a meta analysis of published papers. Why should the effect size be identical in studies with different populations, with different staff administering the treatments, and where the interventions might not even be identical? Thus I would urge the authors to modify their protocol to use a random effects model in all circumstances. Reporting the value of I-squared is useful as a summary measure of between study variability, but the only time I feel it should change the analysis is when its values are very large. In those circumstances, I suggest that the authors plan to investigate possible causes of the large degree of heterogeneity. I note that the authors hint at that in the ‘Data synthesis’ section. and also in the ‘Subgroup analysis’ section. There would be a benefit to the protocol if these 3 sections were harmonised to avoid repetition when possible, with cross-referencing across sections when necessary.
Re: We appreciate and agree with the reviewer’s advice. We have re-written this part according to the reviewer’s suggestion：
① Assessment of heterogeneity
Forest plots will be used to present effect sizes and corresponding 95% confidence limits for each

study. At the same time, the inconsistency will be quantified by calculating the I^2 test.

② Data synthesis
Data synthesis will be performed through RevMan software (V.5.3). The random-effects model will be used to pool the data. When necessary, the possible causes of the large degree of heterogeneity will be analyzed or a subgroup analysis will be conducted. If the heterogeneity is considerable in the included trials, no meta-analysis will be performed.

4. At the start of the section on 'Assessment of heterogeneity', the term forest plot survey is used, and the authors specify a significance level of 0.1 when investigating statistical heterogeneity. I think the authors simply need to say that forest plots will be used to present effect sizes and corresponding 95% confidence limits for each study. The term 'survey' is unsuitable. In view of my suggestion that random effects models should always be used, I think that a significance test for heterogeneity is futile. The random effects model already assumes that there could be different effect sizes and the Null hypothesis of no differences in effect sizes is therefore irrelevant.
Re: We appreciate and agree with the reviewer's advice. We have re-written this part according to the reviewer's suggestion：
Assessment of heterogeneity
Forest plots will be used to present effect sizes and corresponding 95% confidence limits for each study. At the same time, the inconsistency will be quantified by calculating the I^2 test.

5. The authors have clearly thought about the search strategy and I do not have the expertise to comment, other than to mention a typo in Table 1 where, in step 16, 'mamual' should be replaced by 'manual'. I notice, though, that the authors have not included acupressure in their search. Would such a method fall within the scope of this systematic review? Unless the authors have already done so, I would urge them to consult an experienced librarian about the search strategy, as librarians are the experts. Comparing the search strategy in this protocol to that in the protocol of Yin et al., I see that the search for acupuncture studies is far more detailed in the latter study.
Re: We appreciate and agree with the reviewer's advice.
① Thank you very much for pointing out the typo. We are very sorry for our incorrect writing.
② Considering the reviewer's suggestion, we consulted an experienced librarian about the search strategy. Acupressure is not fall within the scope of this systematic review. And we have supplemented the search strategy.
Table 1 Search strategy used in MEDLINE database
#16 acupuncture (MeSH)
#17 (acupuncture or body acupuncture or manual acupuncture or electroacupuncture or electro-acupuncture or acupoint or auricular acupuncture or warm needling): ti,ab

Special thanks to you for your good comments.

Responses to Reviewer 2:
This is a well-designed protocol for a systematic review. I only have a few suggestions for authors.
Re: Thank you for the reviewer's rigorous consideration.

1. The RevMan software has been updated to version 5.4. In future data analysis, this version should be considered.
Re: We appreciate and agree with the reviewer's advice. RevMan software (V.5.4) will be used in this study.

2. Authors mentioned acupuncture details will be corrected based on STRICTA statement. I suggest

author list the detailed information.
Re: We appreciate and agree with the reviewer's comments and advices. We have re-written this part according to the reviewer's suggestion:
Data extraction and management
The acupuncture intervention details (such as the type of acupuncture, number of needle insertions, depths of insertion, selection of points, de-qi, needle stimulation) will be elaborated according to STandards for Reporting Interventions in Clinical Trials of Acupuncture (STRICTA).23

3. Publication bias will be evaluated through a funnel plot according to manuscript. I suggest Egger's test could be used to assess the imbalance.
Re: We appreciate and agree with the reviewer's comments and advices. Considering the reviewer's suggestion, we have modified the manuscript：
Assessment of reporting biases
A funnel plot for observing the reporting biases will be generated when more than ten studies are included.25 Potential reporting biases will be further assessed with Egger's test

4. In addition, meta-regression could be considered for exploring the potential resources of heterogeneity.
Re: We appreciate and agree with the reviewer's comments and advices. We have re-written a part according to the reviewer's suggestion
Other analysis
The use of meta-regression analysis will be considered to explore the potential heterogeneous resources.
Special thanks to you for your good comments.

We appreciate for Editors/Reviewers' warm work earnestly, and hope that the correction will meet with approval.
Special thanks to you for your good comments.

VERSION 2 – REVIEW

REVIEWER	Robin Prescott University of Edinburgh UK
REVIEW RETURNED	04-Dec-2020

GENERAL COMMENTS	I thank the authors for considering my comments on the original draft and for making changes where appropriate. These changes are generally satisfactory. There is one area where there have been some limited improvements, but where greater clarity could be achieved. That is the section on missing data. I think this section will become clearer if the authors start by identifying the kind of information that may be missing in the original trial report and which the reviewers need. Sometimes, the trial report may be expressed in terms other than mean and standard deviation, as is required for using RevMan software. Thus point 2 of the authors in this section is very important, but it might be clearer if the authors expressed this by starting "If continuous outcome data is not expressed as means and standard deviations", rather than expressing this as missing outcome data. The data is not really missing, but it is presented in a way that is unhelpful when performing a meta analysis.

	In relation to point 1, I would totally agree that, if an ITT analysis was performed, that would always be used in the systematic review, but I do not see how that ties in with missing values. It might be relevant if a paper reported both an ITT analysis and a per protocol analysis, but I am unclear how missing data is an alternative. The most common situation where a reviewer might wish to contact an authors if there was something in the paper that was unclear. That might occur, for example, when completing an assessment of bias risk and the risk of bias was unclear from the paper, or asking for means and standard deviations in preference to estimating them from other summary statistics. I am afraid that the first paragraph of this section is unclear and requires further reconsideration. Perhaps the use of missing information, rather than missing data would help the authors, but there is a real need for clarification, as I do not think any reader will understand this section, as it is presently written.

VERSION 2 – AUTHOR RESPONSE

Responses to Reviewer:
I thank the authors for considering my comments on the original draft and for making changes where appropriate. These changes are generally satisfactory. There is one area where there have been some limited improvements, but where greater clarity could be achieved.

That is the section on missing data. I think this section will become clearer if the authors start by identifying the kind of information that may be missing in the original trial report and which the reviewers need. Sometimes, the trial report may be expressed in terms other than mean and standard deviation, as is required for using RevMan software. Thus point 2 of the authors in this section is very important, but it might be clearer if the authors expressed this by starting “If continuous outcome data is not expressed as means and standard deviations”, rather than expressing this as missing outcome data. The data is not really missing, but it is presented in a way that is unhelpful when performing a meta analysis.

In relation to point 1, I would totally agree that, if an ITT analysis was performed, that would always be used in the systematic review, but I do not see how that ties in with missing values. It might be relevant if a paper reported both an ITT analysis and a per protocol analysis, but I am unclear how missing data is an alternative.

The most common situation where a reviewer might wish to contact an authors if there was something in the paper that was unclear. That might occur, for example, when completing an assessment of bias risk and the risk of bias was unclear from the paper, or asking for means and standard deviations in preference to estimating them from other summary statistics.

I am afraid that the first paragraph of this section is unclear and requires further reconsideration. Perhaps the use of missing information, rather than missing data would help the authors, but there is a real need for clarification, as I do not think any reader will understand this section, as it is presently written.

Re: We appreciate and agree with the reviewer’s advice. We have re-written this part according to the

Reviewer's suggestion.

Dealing with missing data
The authors will be contacted by reviewers to obtain the missing information which was not clearly described in each study included. 25 If available, we will preferentially use the data from intention-to-treat analysis.26 If continuous outcome data is not expressed as means and standard deviations, we will try to re-calculate means and standard deviations as the first choice.
At the same time, if feasible, we will conduct a sensitivity analyses to address the potential impact of missing data.27 The potential impact of the effect of missing information on the final findings of the review will be addressed in the discussion.

Special thanks to you for your good comments.

VERSION 3 – REVIEW

REVIEWER	Professor Robin Prescott University of Edinburgh UK
REVIEW RETURNED	17-Dec-2020

GENERAL COMMENTS	I thank the authors for their consideration of my comments. The revised version has been improved and I have no further comments.